HMLV-survey

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List of Acronyms

ML Machine Learning

HMLV High-Mix Low-Volume

IIoT Industrial Internet of Things

IoT Internet of Things

AI Artefitial Intelligence

CPS Cyper-Physical System

MES Manufacturing Execution System

ERP Enterprise Resource Planning

DT Digital Twin

PT Physical Twin

IoDT Internet of Digital Twins (IoDT)

DTaaS Ditial Twin as a Service

MLOps Machine Learning Operations

PLC Programmable Logic Controller

SME Small and Medium-sized Enterprise

OPC UA Open Platform Communications Unified Architecture

OEE Overall Equipment Effectiveness

SPC Statistical Process Control

cobots Collaborative Robots

CNC Computer Numerical Control

SHAP SHapley Additive exPlanations

LIME Local Interpretable Model-agnostic Explanations

DL Deep Learning

co-bots Collaborative Robots

HMI Human Machine Interface

XR Extended Reality

VR Virtual Reality

AR Augmented Reality

MR Mixed Reality

Chapter 1

Introduction

Potential outline for introduction:

- Define High-Mix Low-Volume (HMLV) manufacturing (what it is, why it matters).
- Highlight challenges: Variability, short product life cycles, small batches.
- State survey purpose: e.g., "to review the technological landscape of HMLV manufacturing across industry 3.0, 4.0, and 5.0 paradigms."
- Mention scope: Not a proposal, but a descriptive and comparative mapping of technologies.

Think of the HMLV manufacturing field as a stack comprised of three layers. The foundation, the digital layer, and the human-centric future. With this division, the foundation is what we in this survey present as "Industry 3.0". It can also be called the foundational techniques which has shaped the beginning of HMLV manufacturing. As the digital side made advances, technological innovations were applied to the field of manufacturing, thereby creating the digital layer comprised of Industry 4.0 technologies such as Cyper-Physical System (CPS)s, Internet of Things (IoT), Big Data, Artefitial Intelligence (AI), and Automation. The future might lie in the more human-centric, as a counter-reaction to the non-human paradigm of Industry 4.0. Therefore, the last stack "Human-Centric Future" deals with the future of HMLV manufacturing in the case of Industry 5.0 values, such as a more human-centric, resilient and sustainable industrial paradigm.

In this survey, we aim to collect and present the three different paradigms, and what the cutting-edge techniques within each paradigm are.

1.1 Terminology

This section outlines some of the terminology common to the manufacturing industry, with a particular focus on HMLV. Readers with more experience in the field can easily skip this section.

Shop Floor

The physical area where production takes place. It includes machines, workstations, material handling systems, and personnel. In HMLV, the shop floor is often highly flexible and dynamic due to frequent changes in production orders.

Line

A sequence of machines or workstations through which a product moves during manufacturing. Lines can be dedicated to a single product (low-mix) or reconfigurable to handle multiple products (high-mix).

Cell

A cell is a group of machines, workstations, and operators arranged together to complete a set of related operations or produce a family of products. The idea is to create a self-contained mini-production unit within the shop floor.

Process

A set of operations or steps required to transform raw materials or components into a finished product. Processes can be discrete (individual steps) or continuous (ongoing production).

Unit

A single instance of a product being manufactured. In HMLV, units are often produced in small batches, and each unit may have unique specifications.

Batch

A collection of units processed together, common in low-volume settings.

Job / Order

A customer request specifying product, quantity, and delivery time.

Setup / Changeover

The process of preparing a machine or line to produce a different product, often a focus in HMLV manufacturing.

Component

A part or subassembly that is used to build a unit. Components may come from internal production or external suppliers.

Property

A measurable characteristic of a unit, component, or process, such as dimensions, material hardness, or temperature during processing.

Device

A device is typically also referred to as an edge node, and is any type of electronic device that can measure some property and transmit these values via communication protocols.

Event

A discrete occurrence on the shop floor, such as the start or end of a process, a machine failure, or an inspection.

Additive Manufacturing

A manufacturing process that adds some material to the product, e.g. 3D printing. A definition: "a process of joining materials to make objects from 3D model data, usually layer upon layer, as opposed to subtractive manufacturing methodologies"

Subtractive Manufacturing

A manufacturing process that removes some material from the product, like cutting or milling.

Chapter 2

Methodology

2.1 Search String

("manufacturing" OR "production" OR "resilient") AND ("intelligent" OR "decision support" OR "digital twin" OR "big data" OR "AI" OR "cyber-physical systems" OR "IIoT" OR "human" OR "predictive maintenance")

Chapter 3

Industry 3.0

Some key technologies to consider touching upon in this section:

- Automation & IT systems Programmable Logic Controller (PLC), Computer Numerical Control (CNC), Manufacturing Execution System (MES), Enterprise Resource Planning (ERP).
- Lean manufacturing, Overall Equipment Effectiveness (OEE), Statistical Process Control (SPC), Layout/flow optimization.
- Operations research in scheduling.
- Limitations for HMLV (rigidity, poor adaptability).

3.1 Overall Equipment Effectiveness

One of the most fundamental metrics in industrial manufacturing is OEE. It is a metric that provides insights into the total throughput of individual production devices. The OEE is calculated as a product of three percentage factors: availability, performance, and quality:

$$\text{OEE} = \text{Availability} \times \text{Performance} \times \text{Quality}$$

The availability measures the percentage of time during which the machine is available for executing tasks. Whenever the machine stops, its availability decreases. High availability is achieved when there is minimal downtime.

$$\text{Availability} = \frac{\text{Total Operation Time} - \text{Total Downtime}}{\text{Total Operation Time}}$$

Performance is the ratio between the actual and ideal number of produced units during uptime. High performance is achieved when the actual cycle time is close to the ideal cycle time. The ideal cycle time is typically provided by the machine manufacturer.

$$\text{Performance} = \frac{\text{Ideal Cycle Time} \times \text{Total Units Produced}}{\text{Total Operation Time} - \text{Total Downtime}}$$

The quality is calculated as the number of good units divided by the total number of produced units. If a large number of units produced by the machine are discarded, the Quality will be lowered.

$$\text{Quality} = \frac{\text{Good Units}}{\text{Total Units Produced}}$$

3.2 PackML State

3.3 Simulation and Scheduling

3.4 SPC

3.5 Lean Manufacturing

3.6 Product Line Design / Group Technology

3.7 Layout and Flow Optimization

Chapter 4

Industry 4.0

Some key technologies to consider touching upon in this section (/suggested structure):

- CPSs: Concept, implementations, impact on HMLV.
- IoT & Industrial Connectivity: Sensor networks, Open Platform Communications Unified Architecture (OPC UA), data pipelines.
- Big Data & Analytics: Predictive maintenance, real-time monitoring.
- Automation & Robotics: Reconfigurable automation, co-bots, flexible robotic cells.
- AI/Machine Learning (ML) for Decision Support: Anomaly detection, predictive scheduling, adaptive control.
- Machine Learning Operations (MLOps) in Manufacturing: Deployment challenges (real-time, PLC integration).
- Cloud & Edge Computing: architectures for HMLV environments.

4.1 Cyber-Physical Systems

CPSs are systems in which physical processes are tightly integrated with computational and communication elements. The physical parts (machines, sensors, actuators, infrastrutruce) interact with embedded computation and networking, enabling monitoring, control, and dynamic adaptation. CPS differs from more conventional embedded systems in that they are networked, often large-scale, and involve real-time feedback loops between the physical world and computation. Key early work includes Monostori et al. (2016) [1], which presents the foundational view of CPS / Cyber-Physical Production Systems (CPPS) as central to industry 4.0. Another core is Monostori (2014) "Cyber-physical Production Systems: Roots, Expectations and R&D Challenges" [2].

CPS are seen as essential for enabling the responsiveness, flexibility, and adaptability required in HMLV manufacturing, because they allow systems to sense context, respond in close to real time, and reconfigure or adjust behavior when parts, batches, or processes change.

4.1.1 Role of CPS in Manufacturing

In manufacturing, CPS integrates production equipment with computation and connectivity, enabling continuous monitoring optimization, and reconfiguration. For HMLV contexts, where product variation and small batch sizes are common, CPS make it possible to:

- Collect and analyze sensor data for real-time decision making.
- Enable predictive maintenance and reduce unplanned downtime.

- Support dynamic scheduling and resource allocation under fluctuating demand.
- Improve quality assurance through online defect detection.

Lee et al. (2015) [3] proposed the well-known 5C architecture (connection, conversion, cyber, cognition, configuration), which structures how data from shop-floor devices are transformed into actionable insights. Such architectures illustrate the layered integration of the physical and cyber domains, which ultimately enables self-adaptive production systems.

4.1.2 Framework and Tools

Several frameworks have been proposed to standardize CPS in manufacturing. The Reference Architectural Model Industry 4.0 (RAMI 4.0) [4] provides a three dimensional model linking hierarchy levels, product lifecycle stages, and architecture layers, situating CPS within the broader Industry 4.0 ecosystem.

Research at Aarhus University, led by Larsen et al. [5], has contributed to the development of INTO-CPS, a co-simulation platform for model-based CPS engineering. INTO-CPS supports the design and validation of CPS by integrating multiple modeling tools and enabling virtual experiments across domains. Such platforms exemplify how academic research provides concrete tools for industry to adpot CPS in practice.

4.1.3 Surveys and Applications

Several reviews summarize the state of CPS in manufacturing:

- Dafflon et al. (2021) [6] reviewed 78 CPS studies (2010-2019), identifying common challenges such as interoperability, scalability, and human integration, and highlighting technologies such as IoT, Big Data, and AI.
- Liu et al. (2018) [7] focused on CPS-based smart warehouses, showing how CPS concepts extend to logistics, with applications in sensor-driven monitoring, automated transport, and adaptive storage systems.
- Benzadri et al. (2021) [8] proposed a modeling framework for CPS-based manufacturing systems, demonstrating the practical challenges of applying CPS in industrial settings.

These studies show that CPS research is both broad and maturing, with applications ranging from production lines to logistics and supply chain contexts.

4.1.4 Challenges and Future Directions

Despite progress, several open issues remain:

- Interoperability between heterogeneous machines and software platforms.
- Scalability of CPS solutions for Small and Medium-sized Enterprise (SME)s and HMLV contexts.
- Data quality, security, and privacy in highly connected environments.
- Human-in-the-loop integration, ensuring that CPS augment rather than replace operators (Industry 5.0).

Looking ahead, CPS are increasingly converging with Digital Twin (DT) technologies, enabling tighter synchronization between physical assets and virtual models. In industry 5.0, CPS are expected to evolve toward human-centric and sustainable architectures, where adaptability is balanced with transparency, explainability, and resource efficiency.

4.2 Internet of Things & Industrial Connectivity

The Industrial Internet of Things (IIoT) refers to the networking of physical devices such as sensors, actuators, and machines with computational and communication infrastructures, enabling monitoring, control, and data-driven optimization [9, 3]. The roots of IIoT can be traced to earlier machine-to-machine (M2M) communication in manufacturing, but IIoT expands this concept through standardized communication protocols, cloud and edge connectivity, and scalable integration [10]. Unlike conventional automation systems, IIoT emphasizes large-scale interoperability, continuous data exchange, and the ability to connect heterogeneous devices across organizational and geographic boundaries.

4.2.1 Role in HMLV Manufacturing

For HMLV manufacturing, IIoT provides capabilities that directly address the challenges of variability, short product cycles, and small batch sizes. Real-time sensor networks capture operational data that can be used to rapidly detect process deviations or quality issues when switching between product variants [11]. Connectivity between machines allows production lines to be reconfigured with minimal downtime, supporting the flexibility required for frequent product changes. At the same time, traceability across diverse product runs ensures compliance and quality standards are maintained [12]. In practice, IIoT has been applied in adaptive scheduling, predictive maintenance, and operator decision support systems, all of which are highly relevant to HMLV contexts.

4.2.2 Key Enabling Technologies and Standards

The effectiveness of IIoT depends on several enabling technologies. At the device level, sensor networks and data pipelines provide the foundation for continuous monitoring and feedback loops [10]. At the communication layer, interoperability standards are essential, The OPC UA has become a cornerstone for industrial connectivity, offering secure, structured communication across diverse systems. However, recent studies highlight that performance bottlenecks remain, particularly for latency-sensitive manufacturing processes where rapid response is critical [13].

Beyond protocol-level interoperability, semantic standardization plays a growing role. Industry 4.0 standards are increasingly represented in knowledge graphs and linked-data frameworks, which improve the ability of diverse systems to integrate and interpret shared information [14]. Finally, connectivity infrastructures such as Ethernet, industrial fieldbuses, and emerging 5G and edge computing platforms provide the backbone for IIoT scalability and responsiveness. Low-latency communication is especially important in HMLV manufacturing where production lines may need to adapt in real time to accommodate shifting product configurations [11].

4.2.3 Challenges and Outlook

Despite its promise, IIoT adpotion in HMLV manufacturing faces several challenges. Integrating heterogeneous legacy equipment into IIoT networks is technically demanding and often requires costly retrofitting. Data management is another key barrier: vast volumes of sensor and machine data must be stored, processed, and filtered to extract actionable insights [12]. Security and privacy concerns are heightened as connectivity expands across organizational boundaries, creating new attack surfaces [9]. Organizationally, manufacturers often face cultural and skills-related barriers in adopting IIoT, as the shift requires both technical and managerial adaptation.

Looking forward, advances in edge and 5G infrastructures are expected to mitigate some of the latency and scalability issues. Increasing adoption of semantic interoperability standards will reduce integration complexity, while AI-driven analytics layered on top of IIoT data streams are likely to deliver more powerful decision support. For HMLV manufacturers, these developments could significantly reduce the overhead of frequent product changes and make flexible production systems more cost-effective.

4.3 Big Data & Analytics

Big data refers to the large-scale, complex, and rapidly evolving datasets generated in modern manufacturing environments. These datasets are typically described by the "5Vs": volume, velocity, variety, veracity, and value, which highlight both the opportunities and the challenges of extracting useful knowledge [15]. Analytics denotes the methods and tools that transform these data streams into actionable insights. In manufacturing, this encompasses a continuum ranging from descriptive statistics and dashboards, to predictive modeling, and ultimately prescriptive systems that recommend or even automate decision-making [3, 10]. The combination of Big Data and advanced analytics has been identified as cornerstone of Industry 4.0, providing the intelligence needed to enable self-adaptive and data-driven production systems.

4.3.1 Role in HMLV Manufacturing

The significance of Big Data & Analytics is particularly clear in HMLV production. Unlike mass production, where large amounts of relatively homogeneous data are available, HMLV is characterized by frequent product changes, small batch sizes, and fragmented datasets. This variability makes traditional process optimization difficult, but it also creates opportunities: analytics can uncover patterns across diverse runs, detect anomalies in near real-time, and support decision-making in highly dynamic settings. For example, predictive maintenance models based on vibration or acoustic data can prevent downtime during short production runs [16], while quality analytics can flag deviations before they propagate across multiple product variants [12]. Similarly, adaptive scheduling systems informed by real-time data streams can dynamically allocate resources when demand fluctuates or when product designs are updated [11].

4.3.2 Analytical Approaches and Trends

Several strands of analytics have become central to manufacturing. Descriptive analytics establishes situational awareness, often through visual dashboards and real-time monitoring tools. Predictive analytics leverages statistical and ML methods to forecast equipment failures, yield rates, or process deviations. Prescriptive analytics extends this by recommending or automating corrective actions, for instance rescheduling jobs or adjusting machine settings to maintain throughput. In recent years, there has been growing interest in hybrid methods that combine domain expertise with ML, leading to "physics-informed" models that are more robust under conditions of sparse and variable data [10]. These hybrid approaches are especially attractive for HMLV, where purely data-driven methods often struggle due to limited sample sizes for each product variant.

4.3.3 Challenges and Outlook

Despite clear progress, significant challenges remain in implementing Big Data & Analytics in real-world manufacturing contexts. Data heterogeneity is a persistent barrier, as sensor readings, machine logs, and enterprise-level ERP/MES records must be harmonized to provide meaningful insights [12]. Real-time responsiveness is another challenge: in HMLV settings, decisions must often be made during ongoing production runs, requiring low-latency data pipelines and fast analytical models [11]. Furthermore, analytics must not only be accurate but also interpretable, since operators must be able to trust and act on system outputs. This has created growing interest in explainable AI (XAI) [17] approaches for manufacturing, particularly in contexts where human operator remain central decision-makers. Finally, infrastructure requirements for storing transmitting, and analyzing large volumes of data can be a limiting factor, particularly for SME.

Looking ahead, Big Data & Analytics can be viewed as the bridge between IoT (which provides the raw data streams) and CPSs (which implement adaptive responses). Advances in distributed stream processing, edge analytics, and semantic data integration are expected to reduce latency and improve scalability. At the same time, combining predictive and prescriptive methods promises to support not only efficiency but also resilience, enabling manufacturers to adapt to variability

and disruptions more effectively. In the context of HMLV manufacturing, these developments could significantly reduce the overhead of frequent product changes, making highly flexible production both feasible and economically viable.

4.4 Advanced Robotics & Automation

Automation has long underpinned industrial production, but its nature has shifted dramatically with the transition from rigid mechanization to intelligent, connected systems. Early automation relied on fixed robotic arms, conveyor systems, and PLCs, which were efficient for mass production but inflexible in the face of product variability [18, 2]. In HMLV manufacturing, this rigidity proved especially limiting, as frequent changeovers, short product life cycles, and small batch sizes demand adaptability.

4.4.1 From Fixed Automation to Flexible Systems

Industry 4.0 brought a new paradigm by embedding robotics into broader cyber-physical production systems [1]. Robots are no longer isolated automation units, but interconnected agents within systems of sensing, data exchange, and decision-making. This has enabled a shift toward more flexible, responsive forms of automation.

One important development has been the rise of Collaborative Robots (cobots), which integrate advanced sensing, force control, and safety mechanisms to share workspaces with human operators [19]. Their flexibility makes them particularly relevant to HMLV production, where some assembly or finishing tasks require human dexterity while still benefiting from robotic consistency. IN parallel, reconfigurable robotic systems have merged, emphasizing modular hardware and software architectures that can be adapted to different product types or processes with minimal downtime [20, 21]. While the concept of cobots leans more towards the human-centric ideals of Industry 5.0, the improvement in quality, efficiency and partial automation also makes it relevant to the Industry 4.0 industrial paradigm.

4.4.2 Intelligence and Adaptivity in Automation

Advances in AI and ML extend the scope of robotics beyond repetitive automation. Machine vision, reinforcement learning, and adaptive motion planning now allows robots to handle semi-structured environments and uncertain inputs [22]. These capabilities are especially valuable in HMLV contexts, where production tasks vary and cannot be exhaustively pre-programed. Such developments are closely tied to big data analytics and IoT infrastructures [3, 23], since adaptive automation depends on streams of sensor data and predictive models for decision support.

4.4.3 Networked and Cloud Robotics

The rise of cloud robotics and edge-assisted automation illustrates another step in this evolution. By linking robotic systems through networked infrastructures, capabilities such as shared learning, fleet coordination, or remote monitoring become feasible [24]. These approaches align with the broader Industry 4.0 emphasis on connectivity and data-driven optimization, but also raise questions of latency, security, and interoperability.

4.4.4 Challenges and Future Directions

Despite rapid progress, challenges remain. Advanced robotics often entails high upfront costs, and the return on investment can be difficult to justify in small-back production unless flexibility and reusability are maximized [21]. Human-robot collaboration also introduces ergonomic, trust, and safety considerations that are not yet fully resolved [19]. Integration into wider CPS and IoT infrastructures is further slowed by the lack of universal standards, a theme which is common across Industry 4.0 technologies [2, 23].

Overall, robotics and automation are evolving from rigid, standalone systems into flexible, intelligent, and networked production agents. Their ongoing development is central to achieving the adaptability and responsiveness for HMLV manufacturing, even as integration, cost, and safety challenges remain active areas of research and industrial practice.

4.5 Decision Support

4.5.1 AI-based

4.5.2 Predictive Maintenance

4.5.3 Explainable AI

In [17], the authors define interpretable and explainable AI and review their use cases in manufacturing. The authors describe interpretable AI as methods that are inherently easy to understand due to their data-processing architecture. For instance, random forest methods fall into this category, as it is relatively straightforward to trace a decision through the rules of each tree, albeit a cumbersome task when dealing with many trees. In contrast, explainable AI refers to techniques used to interpret decisions made by complex models such as Deep Learning (DL) models. These techniques often rely on local explanations of individual model outputs, such as SHapley Additive exPlanations (SHAP) values, Grad-CAM, or Local Interpretable Model-agnostic Explanations (LIME).

4.5.4 Identified Areas Where AI can be Used

Before Manufacturing

Product design verification (can then Product be manufactured?). Production planning and optimization (plan the execution in an optimal way). Demand forecasting (how many to produce), raw materials and components forecasting (how much material do we need?).

During Manufacturing

Keep production lines running. Defect free production. Machine health monitoring (predictive maintenance) (maintenance costs a lot). Fault diagnostics (why).

After Manufacturing

QA (check for defects). Testing.

4.5.5 Challenges in AI

- Trusting and understanding black box models.

4.6 MLOps

4.7 Cloud and Edge Computer

4.7.1 VM / Virtualization

Chapter 5

Industry 5.0

Some key technologies to consider touching upon in this section (/suggested structure):

- DTs: Simulation, prediction, continuous synchronization.
- Human-Centricity: Operators in the loop, augmented decision support.
- Resilience & Adaptability: Handling disruptions, supply chain integration.
- Sustainability & Circularity: Energy efficiency, material reuse.
- Trustworthy AI & Explainability: Operators must trust AI outputs.
- Collaborative Robotics: Cobots with human oversight.

5.1 Digital Twins

5.1.1 Digital Twins

The concept of a DT was first proposed in 2003 by Micheal Grieves [25]. Since then, numerous definitions and concepts have been made [26]. A DT can be defined as a digital representation of a physical entity, referred to as the Physical Twin (PT). Both are connected by a bidirectional communication infrastructure, feeding real-time data from PT to DT and control feedback from DT to PT [27]. DT technology is seen as a key technology in Industry 4.0 and Industry 5.0, in the manufacturing lifecycle, and in human-centric manufacturing such as human-robot collaboration. Kritzinger et al. (2018) [28] propose that DT can be categorized in terms of connectivity (manual vs automatic data flow) to the physical entity. Thus, a digital representation can be either a digital model, a digital shadow, or a digital twin, enabling a clear definition of DT technology in its level of integration. DTs has been widely used in the manufacturing product lifecycle. In the design, production, prognostics, and health management areas [29]. The wide applicability of DTs makes them relevant for HMLV manufacturing, where high variety, small batches, and shorter product life cycles require adaptability, flexibility, and efficacy [30].

5.1.2 Case Studies and Architectures

In the integration of DTs in manufacturing processes, several architectures has been suggested, to do this. In[31] they propose that the DT can be incorporated in the design, manufacturing, maintenance, recycling, and scrapping phase, establishing a full life cycle data center for the PT extended with intelligent decision-support methods such as Analytic hierarchy process (AHP), Bayesian estimation, expert systems, D–S reasoning, and AI.

Several case studies have also been made in the field of DTs in manufacturing. Some of which are the following:

- Ren et al. (2022) [31] propose a DT in maintenance model of a locomotive entity for maintenance prediction through predictive models, demonstrating the effectiveness and practical value of DTs in health management maintenance.
- Luo et al. (2020) [32] implement a hybrid predictive maintenance method that combines data-driven and multidomain models within a DT framework, illustrating that DT technology can provide a framework for combining data-driven and physics-based approaches to improve predictive maintenance accuracy.
- Villani et al. (2025) [33] present a human-in-the-loop operator digital twin (ODT), giving the operator insight into the plant's state, while also giving technological agents in the plant insight into the operator's physical, emotional, and mental conditions, demonstrating how DT technology can be exploited to realize human-centric industrial applications. Human-robot collaboration case, such as human-robot collaboration.
- In [34] they present a real-time rescheduling system based on a DT of a shop floor. It reacts to manufacturing job events considering process time, production unit volume, worker, machine, operation, etc. Achieving better rescheduling performance by having more accurate data and a timely response to dynamic events, compared to traditional methods.

5.1.3 Enabling Technologies

To effectively implement DTs in manufacturing, several enabling technologies are required. At its foundation, IoT/IIoT (sensors and actuators) and data acquisition networks enables the ability to capture data from machines and processes from the physical system used by the DT. As well as storage technology to store the collected data [25, 35]. To communicate in heterogeneous environments, communication protocols and interoperability standards ensure reliable and secure data transmission between physical and virtual layers [36]. Cloud and edge computing, together with big data analytics and AI, provide a computational infrastructure to process and analyze the collected data [25, 37]. Finally, data-driven and physics-based models, along with visualization and human-machine interfaces, enable monitoring, decision-making, and interaction with the physical system [38].

5.1.4 Surveys

On the topic of DTs, there exist several surveys on DTs in manufacturing, setting its current state:

- This survey first describes the concept of DT and its misconceptions. It then discusses the challenges of IoT/IIoT, data analytics, and DTs in today's context. Followed by an identification of their enabling technologies. Lastly, it ends with a categorical review of current research on IoT/IIoT and data analytics with a focus on DT publications. Reviewing 177 articles from 2015 to 2019, categorizing them into three areas: Manufacturing, Healthcare, and smart city [35].
- [36] reviews 92 publications on DTs from 2009 to 2018, focusing on identifying thematic areas, future directions and gaps in research. It highlights gaps such as the lack of attention to data ownership and integration between virtual entities. A key finding is the absence of studies that provide quantifiable evidence of the benefits of a DT approach compared to existing processes and systems, which can hinder industrial adoption. Furthermore, it also highlight that most litterateur is concentrated on specific phases of the life-cycle, with early stages being less explored. Finally, it compares related research domains within the context of DTs.
- Wang et al. (2023) [37] presents the concept of Internet of Digital Twins (IoDT) (IoDT), its architecture, enabling technologies, and security/privacy issues. IoDT is an interconnected system of a number of DTs and Physical entities synchronized through intra-twin semantic

communication. In the survey, they highlight their contribution compared to other surveys on the DT topic from 2019 to 2023.

5.1.5 DTaaS

While DT technology has an influential future, its prolonged life depends on practical implementations, value-added services, and product realization. Studies on Ditial Twin as a Service (DTaaS) have been approached from two angles. (i) a conceptual reference architecture model as a means to realize mass individualization, and (ii) a platform-oriented implementation.

In [39], Aheleroff et al. (2021) propose a DT reference architecture model in Industry 4.0. Inspired by RAMI 4.0 and the DIKW model (Data, Information, Knowledge, and Wisdom), they present a layered reference model, which breaks down the complex integration of DTs in Industry into manageable packages with a focus on the DT Layers, level of integration, and agile methodology. While also being a common understanding and reference point among contributors.

From a platform and implementation perspective, Talasila et al. (2022) [40] present an architecture and concrete implementation of DTaaS. Focusing on the full DT lifecycle (Create, Execute, Save, Analyse, Evolve, Terminate) and the reusability of DT assets, categorizing the DT assets into Data, Mode, Function, and Tool, thereby lowering the stepping stone for adoption for non-technical users.

5.1.6 What-if (offline)

By having a system consisting of PT and DT, one can perform a What-if simulation. What-if simulation is done by having historical data, which is used to simulate an event using different parameters before applying the change to the actual physical object. In [41], they illustrate how what-if can be used to determine a suitable heating power-on time before performing a heater replacement to account for the lack of heating while the replacement happens. This allows the operator to test different power-on times to ensure the avoidance of being in an unwanted state. The ability to create what-if simulations in HMLV industries can reduce the design-to-manufacturing phase, thereby giving a faster response to new demands.

5.1.7 Digital Twin Challenges and Future Directions

While the DT technology has gained much attention through the years since its conception in 2003 by Micheal Grieves [25], challenges still exists. A lack of standardization, frameworks, and reference models that unify the process of creating and implementing DTs in the manufacturing industry. A unified reference model for implementation has been proposed [39]; however, it has not achieved widespread adoption, highlighting the need for reference models that satisfy domain-specific requirements. A common DT definition isn't fully agreed upon in research, while most use the term DT, a common understanding and definition are still missing. DTs rely on real-time two-way communication between he physical and DT to ensure that the DT properly reflects the PT. Achieving this is challenging due to the spatio-temporal resolution of sensor data, latency in communication, and the large volume and variety of data that is generated citeDigital Twin, Challenges and Enablers From a Modeling Perspective. To alleviate this, research into more robust fusion algorithms to ensure better synchronization of physical and DTs is needed. While also attending to the computational burden to perform complex computations on large volumes of data at high velocity, citePredictive maintenance using digital twins: A systematic literature review poses a challenge. This could be handled using compute technologies such as edge computing and cloud computing.

However, obstacles remain how detailed a DT can or should be. Aheleroff et a. [39] highlight challenges such as knowledge isolation, understanding the physical world, and the computational power needed for simulating the sheer number of states a system can have under many parameter configurations. They propose three dimensions to assess this challenge: Integration level (How

specific will the DT integration be?), Interval (How detailed and accurate should the DT be?), and Intricacy (How much resource is required to create a DT?).

The DT fidelity level describes how closely it can reflect the PT. The higher the fidelity is, the more it resembles the PT. Currently, no study implements a high-fidelity model and often just uses a subset of the parameters. Implementing high-fidelity models in DTs is not currently achievable as they require high network speeds and computational processing power [36], which is not doable in all applications. Such as real-time critical systems. Perceiving the benefits of DTs is not trivial, and there is a lack of research in validation and quantification against existing processes and systems to preemptively assess the benefits as well as determine whether the achieved performance stems from the DT or its enabling technologies [36, 42].

5.2 Human Centric

A key distinction between Industry 4.0 and Industry 5.0 is the inclusion of the human as a central part of the manufacturing process. While Industry 4.0 emphasizes mass production, mass productivity, and performance through automation, often to eliminate or minimize the need for human interventions, industry 5.0, takes a human-centric approach, striving to achieve the most effective collaboration between humans and machines or robots, with the humans well-being in focus. In this paradigm, robots and automated systems are assigned with repetitive and labor-intensive tasks, whereas humans serves as decision makers contributing with critical thinking, creativity, problem-solving skills and their innovative mindset. Such collaboration is particularly valuable in complex tasks and non-routine tasks, that are not easily automated. [43]. In HMLV, this human-centric approach allows for better adaptability and flexibility to product variations and disruptions.

5.2.1 Enabling technologies

To enable a human-centric approach, several technologies can be utilized, including collaborative robots Collaborative Robots (co-bots), AI, ML, Human Machine Interface Human Machine Interface (HMI), Extended Reality (XR), and DT. Co-bots in industrial settings are developed to collaborate with factory workers, assisting them and alleviating them from repetitive tasks, while ensuring a safe and productive environment. Moreover, AI and ML can be employed for decision making, by supporting applications such as with predictive maintenance, quality control, and supply chain optimization.

HMI plays an important role in enabling human understanding and interaction with machinery in manufacturing settings. Through HMIs, operators can see the current state of the machine, and see real-time information about performance and production status. Beyond 2D screens, other enabling technologies such as: Extended Reality XR technologies, such as Virtual Reality Virtual Reality (VR), Augmented Reality Augmented Reality (AR), and Mixed Reality Mixed Reality (MR) present a more immersive way for humans to interact with machines. These technologies can be used by the operator in various contexts, including production line monitoring, remote assisting, virtual assembly and maintenance instructions, navigation inside and outside the manufacturing facility and training of new employees/operators. [44]. In [45] they demonstrate how safety in human–machine interaction can be enhanced using AR, digital twins, and AI. They propose an integrated mixed reality system for safety-aware human–robot collaboration, where DL and DT track workspace safety and assist operators in real time through smart MR glasses.

I think this might be a great idea but you would need to identify a good actual use case or do you already have one?

5.3 Resilience and Adaptability

5.4 Sustainability and Circularity

5.5 Ethics and Trustworthy AI

5.6 Co-bots

Chapter 6

Conclusion

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